

Faculty of Electrical Engineering
Banja Luka
Internet programming

Terms of reference December 2025

Create a simple system for managing student internships. The system consists of several parts which are described below:

Spring Boot *backend* application and database

The system should have one database for all applications. Independently define which data will be saved in the system. Functionalities of the SpringBoot application that will be used in other applications are implemented as RESTful services.

It is necessary to implement functionality that automatically creates recommendations for available internships for students. Students are shown a list of a maximum of 5 internships that suit them best based on their interests and skills. Use an arbitrary AI API to create recommendations for practices. The backend application needs to send a text description of the specific student's CV and a description of all internships to the AI API. Data on CV and internships are in the database and are entered into the system through other *web* applications. It is allowed to make one or more requests to the AI system to obtain results, depending on the number of tokens used. The AI for each practice returns a value in the interval from 0 to 1 that represents the degree of compatibility of the practice for that student and a short explanation. *The backend* application creates a ranking list from the received data and returns that data as a result. It is possible to create a free Gemini account through the student email and integrate the use of AI models into the SpringBoot application. Usage example: <https://ai.google.dev/gemini-api/docs/quickstart#java>.

Create an RSS feed that returns information about practices. There is no need to implement a separate client or integrate the RSS feed in other ways.

It is allowed to add data that are useful for the system's work processes and are not defined in the task text. Save all parameters in *the properties* file.

Application for students

The application should have a login page. Student accounts should be in the database and no registration option should be created. After applying, students have the following options:

- CV creation: enable input of general data and *upload* of images, data on education and training, previous work experience, practices, skills and interests. Use the Europass CV format.
- Overview of available internships and application: enable search and filtering by companies and technologies. Create a tabular view and a separate page with details.
- Overview of AI recommended practices: create a display of results as cards (UI *card* elements).
- Completing the work diary: involves entering the student's activities on the weekly level.

On all pages for viewing data, it is necessary to enable search by at least values from one column. Create pagination (pagination) of tables, or **virtual scroll** for lists.

The application must have a uniform appearance on all pages. Use Angular or React for development. It is allowed to use ready-made libraries such as Bootstrap or Material.

Application for colleges

There is a registration form on the home page. If the application is successful, a page opens with options:

- Management of companies: create a tabular view, add a new company and account to login to the application for companies, activate and deactivate company accounts.
- Overview of all internship ads.
- Management of student data (CRUD). Enable the upload of data about students from CSV file.
- Monitoring of students' work: review of work logs, feedback received from companies, student evaluation.

Application implementation must be using JSP M2. This application can use the RESTful services of the Spring Boot application. It is allowed to use ready-made libraries for user interface design. The appearance of the application should be primarily adapted to devices with small screens.

Application for companies

Only faculty-approved companies have access to this application. If the login is successful, a page with options opens:

- Management of practices (CRUD): enable entry of description of practice, used technologies, maintenance period and additional conditions and restrictions.
- Review of students applying for internships: Companies can review CVs created by students and have options to accept and reject. Create option to download CV in PDF format.
- Evaluation of students who are going to practice: review of the work diary, input of comments and similar.
- Changing the access password.

Implement the application using JSP. This application can use the RESTful services of the Spring Boot application. It is allowed to use ready-made libraries for user interface design.

The appearance of the application should be primarily adapted to devices with small screens.

Notes:

- The project task is done individually. • When submitting the project assignment, it is necessary to submit the complete source code and model data. It is necessary to define the test data.
- The use of code generators, except those which were used in laboratory exercises or lectures.

- The user interface should be organized in a uniform way, ie, everything pages should have a similar layout and intuitive controls and display.
- Each type of server validation needs to be implemented as efficiently as possible. Mandatory do client validation.
- It is not allowed to use stored procedures, functions and triggers. Therefore, the database must not have any logic other than the defined tables and the restrictions that apply between them.
- This assignment is valid until a new assignment is published.